DOMINIQUE LOYER

PhD Candidate in AI (Cognitive Informatics) | AI Researcher

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ISBN Prefixes: 978-1-0699945, 978-1-0699904

Professional Summary

Doctoral researcher in AI (cognitive informatics) specializing in explainable AI (xAI), neuro-symbolic systems, and information credibility verification. Published researcher with 13+ peer-reviewed works addressing algorithmic governance, transparency, and accountability. Research combines ontological frameworks with machine learning for interpretable AI systems. Active open-source contributor with software on GitHub and Zenodo.

Education

PhD in AI (Cognitive Informatics) *2024–2026*
Université du Québec à Montréal (UQAM) Montréal, QC
Dissertation defense: April 3, 2026 | **Status:** Examen de Synthèse Doctoral completed
Dissertation: Hybrid System for Information Credibility Evaluation using Neuro-Symbolic AI
Affiliation: Institut des Sciences Cognitives (ISC)

Master of Artificial Intelligence *2014–2016*
Laval University Québec, QC

Master of Business Administration (MBA) *2014–2015*
Laval University Québec, QC

Bachelor of Finance *2001–2004*
HEC Montréal Montréal, QC

Other Training

Google Advanced Data Analytics Professional Certificate *2023*
Coursera, Online 6 months

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Coursera, Online 6 months

Deep Learning Summer School Camp *2019*
Mila, Montréal Intensive training

Professional Experience

AI Consultant *2013–Present*
Sherbrooke Informatique Inc. Montréal, QC

- 13 years consulting in AI, data analytics, and machine learning
- Python: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Scikit-Learn, PyTorch, TensorFlow, Keras
- Additional: R, SQL, Bash, PowerShell, JavaScript, C, SAS; Tableau, Power BI; Spark, Hadoop

Research Interests & Expertise

- Algorithmic Systems & Governance: Power, opacity, accountability in AI systems
- Explainable AI (xAI): Interpretability and transparency
- Neuro-Symbolic AI: Hybrid systems combining logic with neural networks
- Information Credibility: Fact-checking and misinformation detection

- AI Ethics & Accountability: Responsible AI frameworks

Publications & Research Outputs

PhD Synthesis Examination (2025)

Le Léviathan Algorithmique: pouvoir, opacité et responsabilité
The Algorithmic Leviathan: Power, Opacity, and Responsibility in the Age of AI
ISBN: 978-1-0699904-7-1 | DOI: 10.13140/RG.2.2.28011.20002

Research Articles (2025–2026)

Twelve Lessons in Machine Learning: A Survey of Complex Problem Modeling
Based on Domingos's "A Few Useful Things to Know About Machine Learning"
ISBN: 978-1-0699945-7-8 | DOI: 10.13140/RG.2.2.27189.54244
Journal article (unpublished), January 2026

Hybrid System for Information Credibility Verification
ISBN: 978-1-0699945-0-9 | DOI: 10.13140/rg.2.2.36348.24961

Information Verification Ontology
ISBN: 978-1-0699945-1-6 | DOI: 10.13140/RG.2.2.22926.47680

TREC Information Retrieval Evaluation
ISBN: 978-1-0699904-8-8 | DOI: 10.13140/RG.2.2.22608.62721

Survey: Yoav Shoham's Knowledge Representation
ISBN: 978-1-0699904-6-4 | DOI: 10.13140/RG.2.2.13627.24800

Neural Machine Translation (English-Russian)
ISBN: 978-1-0699904-9-5 | DOI: 10.13140/RG.2.2.17980.55687

Query Expansion on TREC AP 88-90
ISBN: 978-1-0699904-5-7 | DOI: 10.13140/RG.2.2.13481.57120

Research Software & Open Source

- **FactCheckingSys** – github.com/DominiqueLoyer/FactCheckingSys
- **sysCRED** – github.com/DominiqueLoyer/sysCRED
- **TREC Framework** – Zenodo DOI: 10.5281/zenodo.17943817
- **ISC Platform** – plateforme-isc.uqam.ca (Full-stack development)

Technical Skills

- **Programming:** Python, R, Bash, LaTeX, SQL, JavaScript, C, SAS
- **AI & ML:** Neural Networks, NLP, Neuro-Symbolic AI, Ontologies
- **Tools:** Git, GitHub, Tableau, Power BI, Spark, Hadoop

Academic & Institutional Involvement

Institut des Sciences Cognitives (ISC) *2024–Present*
Université du Québec à Montréal (UQAM) Montréal, QC

- Full-stack developer: ISC platform
- ISC Information Systems Collective (ISC-2025) member

Scholarly Networks *2022–Present*

- **ORCID:** 0009-0003-9713-7109

- **GitHub:** github.com/DominiqueLoyer
- ResearchGate, Google Scholar

Languages

- **French:** Native proficiency
- **English:** Professional working proficiency

Last updated: January 25, 2026 | ORCID: 0009-0003-9713-7109 | dominiqueloyer.github.io